FEU Institute of Technology
COLLEGE OF ENGINEERING • COLLEGE OF COMPUTER STUDIES

Computer System and Architecture

Machine Problem

MIDTERM

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Data Performed	Date Submitted

List the steps that you perform to come up with the program.
Steps Performed to Develop the Program

Environment Setup & Architecture Selection: Configured the target environment for a Linux 32-bit (elf32) binary using NASM and the GNU Linker (ld), establishing direct x86 Linux system calls (sys_read, sys_write, sys_ioctl, and sys_exit) via interrupt 0x80.

Terminal Mode Configuration: Implemented terminal I/O control using sys_ioctl (TCGETS/TCSETS) to switch the terminal from canonical mode to raw mode, disabling local ECHO and ICANON. This prevents keystrokes from automatically printing twice and stops newline characters from shifting screen rows.

Alternate Screen Buffer & ANSI Cursor Positioning: Designed screen management using standard ANSI escape codes (ESC [?1049h and ESC [?1049l) to provide an isolated full-screen buffer, combined with dynamic cursor movement routines (ESC [<row> ; <col> H) to print interface components at fixed coordinates.

Scrapbook Layout & ASCII Art Implementation: Drafted a structured multi-section layout containing text banners, decorative box borders, an ASCII rabbit avatar, section dividers, and decorative icons stored as null-terminated byte strings in the .data segment.

Interactive Input Handler with Boundary Checking: Developed an interactive string input routine (read_line) that reads characters sequentially from stdin, respects predefined maximum character limits to avoid box overflow, processes Backspace (ASCII 8 and 127) visually, and commits inputs upon pressing Enter (\n or \r).

Graceful Exit & State Restoration: Structured the termination routine to restore the original termios configuration, switch back to the main terminal buffer, reposition the cursor, and exit cleanly via sys_exit (eax = 1).

Paste the source code and explain the functionalities and take a screenshot of the final output.

```
section .data
    init_term    db 27, '[?1049h', 27, '[2J', 27, '[H', 0
    len_init     equ $ - init_term

    exit_term    db 27, '[?1049l', 0
    len_exit     equ $ - exit_term
```

```
    ; --- Top Scrapbook Title Banner ---
    title_t1 db ' _    _                                          _
_', 0
    title_t2 db '/ \  |_) _       |_   |\/| _   (  _  _ _  _  |_) _
/ \ |/ ', 0
    title_t3 db '\_/  |_) (_) |_| |_   |  |(/_  _) (_ | (_||_) |_)
(_)\_)|  \', 0
    title_t4 db '                                          |
', 0

    ; --- Top Right Bunny Card ---
    card_01  db '+------------------------+', 0
    card_02  db '| *                    * |', 0
    card_03  db '|      (\          /)     |', 0
    card_04  db '|     (  \        /  ) * |', 0
    card_05  db '|  *   (  \_---_/  )      |', 0
    card_06  db '|       /  o    o  \    * |', 0
    card_07  db '|      |      ^      |    |', 0
    card_08  db '|  *    \    ---    /     |', 0
    card_09  db '|      ( (")---(") )   * |', 0
    card_10  db '+------------------------+', 0

    ; --- About Me Section ---
    abt_t1   db '+--[ ABOUT ME
]---------------------------------------+', 0
    abt_name db '| Name:
|', 0
    abt_mail db '| Email:
|', 0
    abt_web  db '| Blog/Web:
|', 0
    abt_b1   db
'+-------------------------------------------------------+', 0

    ; --- My First Section with Trophies ---
```

```
    fst_t1    db '+--[ MY FIRST
]-------------------------------------------------------------+',
0
    fst_ach   db '| First big achievement:
\_/  \_/  |', 0
    fst_rsk   db '| First risk I ever took:
\ /  \ /  |', 0
    fst_hap   db '| First time I felt completely happy:
(o-~)(o-~) |', 0
    fst_b1    db
'+-------------------------------------------------------------------
-----------+', 0

    ; --- Widened My Faves Column ---
    fav_t1    db '+--[ MY FAVES ]----------------------+', 0
    fav_c1    db '| * Color/s:                          |', 0
    fav_c2    db '| * Perfume:                          |', 0
    fav_c3    db '| * Music:                      [STAR]|', 0
    fav_c4    db '| * Singer/s:                         |', 0
    fav_c5    db '| * Song:                             |', 0
    fav_c6    db '| * Food:                       d b   |', 0
    fav_b1    db '+-------------------------------------+', 0

    ; --- Widened Hobbies Column ---
    hob_t1    db '+--[ HOBBIES ]-----------------------+', 0
    hob_c1    db '| Weekend:                            |', 0
    hob_c2    db '| TV Show:                            |', 0
    hob_c3    db '| Movie:                        -[=]-|', 0
    hob_c4    db '| Book:                         |(o)||', 0
    hob_c5    db '| Celebs:                        - -  |', 0
    hob_c6    db '| Role model:                         |', 0
    hob_b1    db '+-------------------------------------+', 0

    ; --- Ambition & Motto Section ---
    amb_t1    db '+--[ AMBITION & MOTTO
]-----------------------------------------------------+', 0
```

```
    amb_row1 db '| AMBITION:
|', 0
    amb_row2 db '| MOTTO:
|', 0
    amb_b1   db
'+-------------------------------------------------------------------
-----------+', 0

    ; --- Footer Bar ---
    ftr_bar  db '(^.^)======== Type info & press Enter per field.
Last field exits ========(^.^)', 0

    backspace_seq db 8, ' ', 8

section .bss
    char_in       resb 1
    cursor_seq    resb 16
    orig_termios  resb 64
    raw_termios   resb 64

section .text
    global _start

%macro print_at 3
    mov eax, %1
    mov ebx, %2
    call move_cursor

    mov esi, %3
    call print_string
%endmacro

%macro read_at 3
    mov eax, %1
    mov ebx, %2
    call move_cursor
```

```
    mov ecx, %3
    call read_line
%endmacro

_start:
    ; 1. Enter non-canonical raw mode (disable ICANON and ECHO)
    mov eax, 54           ; sys_ioctl
    mov ebx, 0            ; stdin
    mov ecx, 0x5401       ; TCGETS
    mov edx, orig_termios
    int 0x80

    mov esi, orig_termios
    mov edi, raw_termios
    mov ecx, 16
    rep movsd

    and dword [raw_termios + 12], ~(0x0002 | 0x0008)
    mov byte [raw_termios + 23], 1
    mov byte [raw_termios + 22], 0

    mov eax, 54
    mov ebx, 0
    mov ecx, 0x5402       ; TCSETS
    mov edx, raw_termios
    int 0x80

    ; 2. Switch to alternate screen buffer
    mov eax, 4
    mov ebx, 1
    mov ecx, init_term
    mov edx, len_init
    int 0x80

    ; 3. Draw Title Banner
```

```
    print_at 1, 3, title_t1
    print_at 2, 3, title_t2
    print_at 3, 3, title_t3
    print_at 4, 3, title_t4

    ; 4. Draw Bunny Card
    print_at 1, 56, card_01
    print_at 2, 56, card_02
    print_at 3, 56, card_03
    print_at 4, 56, card_04
    print_at 5, 56, card_05
    print_at 6, 56, card_06
    print_at 7, 56, card_07
    print_at 8, 56, card_08
    print_at 9, 56, card_09
    print_at 10, 56, card_10

    ; 5. Draw About Me Box
    print_at 6, 2, abt_t1
    print_at 7, 2, abt_name
    print_at 8, 2, abt_mail
    print_at 9, 2, abt_web
    print_at 10, 2, abt_b1

    ; 6. Draw My First Box
    print_at 11, 2, fst_t1
    print_at 12, 2, fst_ach
    print_at 13, 2, fst_rsk
    print_at 14, 2, fst_hap
    print_at 15, 2, fst_b1

    ; 7. Draw Faves & Hobbies
    print_at 16, 2, fav_t1
    print_at 17, 2, fav_c1
    print_at 18, 2, fav_c2
    print_at 19, 2, fav_c3
```

```
    print_at 20, 2, fav_c4
    print_at 21, 2, fav_c5
    print_at 22, 2, fav_c6
    print_at 23, 2, fav_b1

    print_at 16, 41, hob_t1
    print_at 17, 41, hob_c1
    print_at 18, 41, hob_c2
    print_at 19, 41, hob_c3
    print_at 20, 41, hob_c4
    print_at 21, 41, hob_c5
    print_at 22, 41, hob_c6
    print_at 23, 41, hob_b1

    ; 8. Draw Ambition & Motto
    print_at 24, 2, amb_t1
    print_at 25, 2, amb_row1
    print_at 26, 2, amb_row2
    print_at 27, 2, amb_b1

    ; 9. Draw Footer
    print_at 28, 2, ftr_bar

    ; --- Expanded Field Inputs (row, col, max_len) ---
    read_at 7, 10, 46     ; Name (was 30)
    read_at 8, 11, 45     ; Email (was 26)
    read_at 9, 14, 42     ; Blog/Web (was 22)

    read_at 12, 27, 40    ; First achievement (was 30)
    read_at 13, 28, 39    ; First risk (was 29)
    read_at 14, 39, 28    ; First happy (was 18)

    read_at 17, 15, 23    ; Color (was 14)
    read_at 18, 15, 23    ; Perfume (was 14)
    read_at 19, 15, 17    ; Music (was 14)
    read_at 20, 16, 22    ; Singer (was 15)
```

```
    read_at 21, 12, 26    ; Song (was 20)
    read_at 22, 12, 22    ; Food (was 20)

    read_at 17, 52, 26    ; Weekend (was 18)
    read_at 18, 52, 26    ; TV Show (was 18)
    read_at 19, 50, 21    ; Movie (was 16)
    read_at 20, 49, 22    ; Book (was 17)
    read_at 21, 51, 20    ; Celebs (was 15)
    read_at 22, 55, 23    ; Role model (was 16)

    read_at 25, 14, 64    ; Ambition (was 60)
    read_at 26, 14, 64    ; Motto (was 60)

    ; Restore terminal modes & exit
    mov eax, 54
    mov ebx, 0
    mov ecx, 0x5402
    mov edx, orig_termios
    int 0x80

    mov eax, 4
    mov ebx, 1
    mov ecx, exit_term
    mov edx, len_exit
    int 0x80

    mov eax, 1
    xor ebx, ebx
    int 0x80

move_cursor:
    push edi
    push edx
    push ecx
    push ebx
    push eax
```

```
    mov byte [cursor_seq], 27
    mov byte [cursor_seq+1], '['
    mov edi, cursor_seq + 2

    call append_num

    mov byte [edi], ';'
    inc edi

    mov eax, [esp + 4]
    call append_num

    mov byte [edi], 'H'
    inc edi

    mov edx, edi
    sub edx, cursor_seq
    mov ecx, cursor_seq
    mov ebx, 1
    mov eax, 4
    int 0x80

    pop eax
    pop ebx
    pop ecx
    pop edx
    pop edi
    ret

read_line:
    push ebp
    mov ebp, esp
    push edi
    push ebx
    push ecx
```

```
    xor edi, edi

.read_char:
    push ecx
    mov eax, 3
    mov ebx, 0
    mov ecx, char_in
    mov edx, 1
    int 0x80
    pop ecx

    cmp eax, 0
    jle .done

    mov al, [char_in]
    cmp al, 10
    je .done
    cmp al, 13
    je .done

    cmp al, 8
    je .handle_backspace
    cmp al, 127
    je .handle_backspace

    cmp al, 32
    jl .read_char

    cmp edi, ecx
    jge .read_char

    push ecx
    mov eax, 4
    mov ebx, 1
    mov ecx, char_in
```

```
    mov edx, 1
    int 0x80
    pop ecx

    inc edi
    jmp .read_char

.handle_backspace:
    test edi, edi
    jz .read_char

    push ecx
    mov eax, 4
    mov ebx, 1
    mov ecx, backspace_seq
    mov edx, 3
    int 0x80
    pop ecx

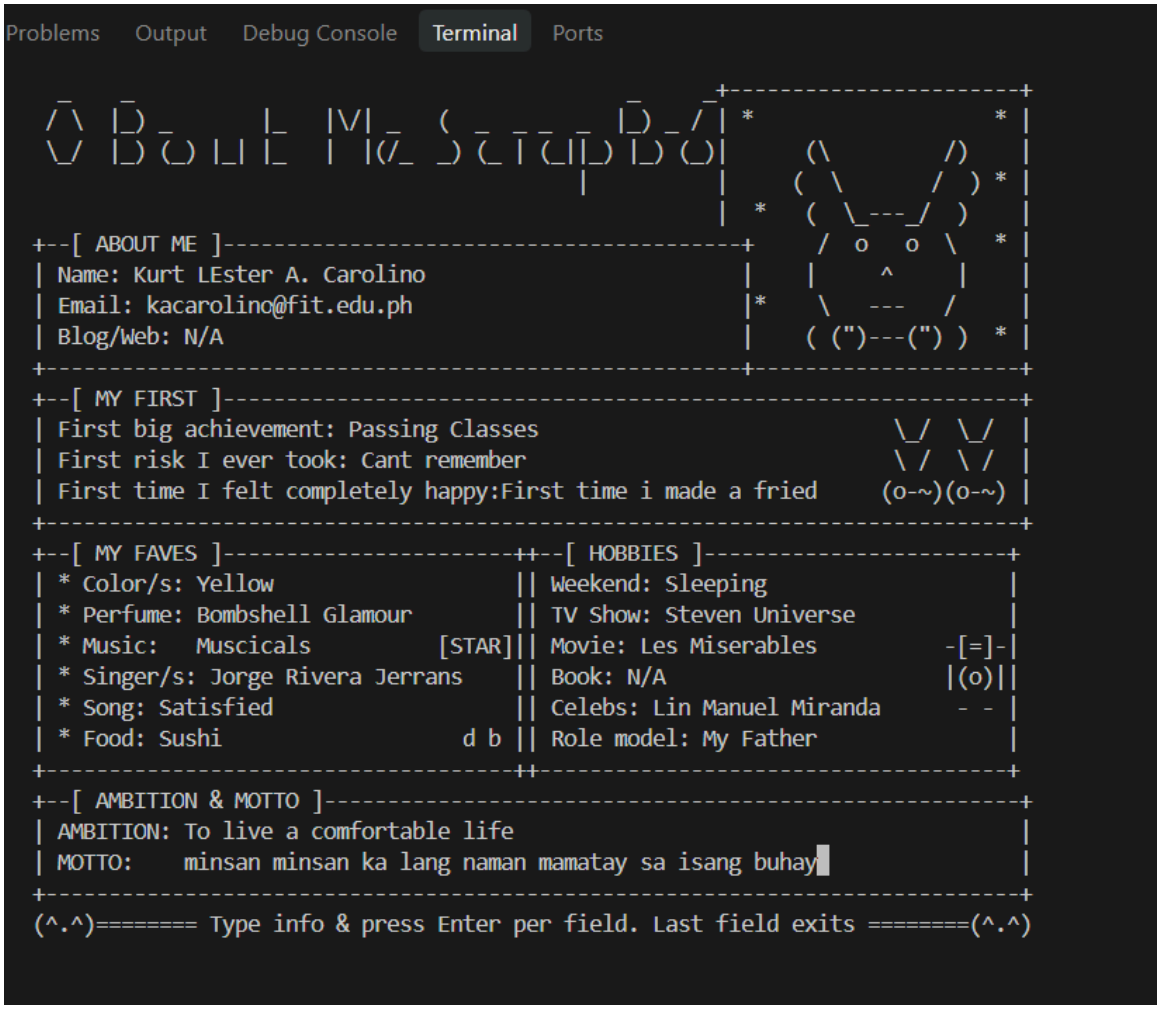
    dec edi
    jmp .read_char

.done:
    pop ecx
    pop ebx
    pop edi
    mov esp, ebp
    pop ebp
    ret

append_num:
    cmp eax, 10
    jl .single_digit
    mov dl, 10
    div dl
    add al, '0'
```

```
    mov [edi], al
    inc edi
    mov al, ah
.single_digit:
    add al, '0'
    mov [edi], al
    inc edi
    ret

print_string:
    push esi
    mov ecx, esi
    xor edx, edx
.scan:
    cmp byte [esi + edx], 0
    je .done
    inc edx
    jmp .scan
.done:
    pop ecx
    mov eax, 4
    mov ebx, 1
    int 0x80
    ret
```


Problems
Output
Debug Console
Terminal
Ports
+--[ABOUT ME]
| Name: Kurt LEster A. Carolino
| Email: kacarolino@fit.edu.ph
| Blog/Web: N/A
+--[MY FIRST]
| First big achievement: Passing Classes
| First risk I ever took: Cant remember
| First time I felt completely happy:First time i made a fried
+--[MY FAVES]
| * Color/s: Yellow
| * Perfume: Bombshell Glamour
| * Music: Muscicals [STAR]
| * Singer/s: Jorge Rivera Jerrans
| * Song: Satisfied
| * Food: Sushi
+--[HOBBIES]
| Weekend: Sleeping
| TV Show: Steven Universe
| Movie: Les Miserables
| Book: N/A
| Celebs: Lin Manuel Miranda
| Role model: My Father
+--[AMBITION & MOTTO]
| AMBITION: To live a comfortable life
| MOTTO: minsan minsan ka lang naman mamatay sa isang buhay
(^.^)======== Type info & press Enter per field. Last field exits ========(^.^)